Assignment 3:

Implementing a Chess-Playing Agent Using Minimax and Alpha-Beta Pruning

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Chess Engine

- Our SearchEngine uses Negamax which is a compact form of minimax that relies on max(a, b) = -min(-a, -b). With this version, the one function plays both sides and every position is scored from the point of view of the player to move and each recursive call negates the child's score. A good move for one side is automatically a bad move for the other (zero-sum).

- Root search (search). It loops over every legal move, pushes the move, and calls minimax on the resulting position at depth - 1. It negates the returned value, because the child is scored from the opponent's point of view. It keeps the move with the highest value and pops the move to restore the board. Because the moves are shuffled first, equally good moves are picked at random, so the agents don't repeat the same game.

- For the Alpha-beta pruning, alpha is the best score the player to move can already guarantee and beta is the best score the opponent will allow. On each recursive call the window flips to (-beta, -alpha). When alpha >= beta, the opponent would never let the game reach this position, so the remaining moves are skipped. Attempting moves that would capture an enemy piece first tends to find strong moves early, which causes more early prunes.

The Agents

The two agents, the Materialist and the Attacker agents both use the same search and minimax functions. Their differences lie in their evaluation strategy functions, in which either agent prioritizes different positions and moves.

The Materialist

The Materialist agent has more of a greedy approach. It scores a position almost entirely by piece count, using full piece values.

- It rewards winning an enemy piece the side to move can capture for free and penalises leaving its own pieces where they can be taken.
- The material lead gets a bonus that grows as pieces leave the board, so it simplifies toward an easy win.
- It ignores where pieces stand, pawn structure and king safety. It plays safe, practical chess and rarely gives anything away.

The Attacker

The Attacker agent has an aggressive approach. It values material at only 85%, so it is willing to give up pieces for a higher position. Most of its score comes from pressure on the enemy king.

- Knights, bishops, rooks and queens get a bonus for being close to the enemy king.
- Pawns get a bonus for marching up the board.
- It is penalised for a broken pawn shield or enemy closeness to its own king.
- Moves that result in a check earns a bonus.

Match Report for Agent vs Agent

Results found in ‘match_log.txt’

Using the Match Harness (match_harness.py) the Materialist and Attacker agents played 10 games against the other at search depth of 2 for both agents, alternating colours for each game.

	Attacker wins	**Materialist wins**
Attacker as White (5 games)	3	2
Materialist as White (5 games)	4	1
Total	7	3

Every game ended in checkmate. The average game lasted about 199 plies (roughly 100 moves).

Analysis

The Attacker’s wins averaged about 174 plies, compared with 258 for the Materialist’s wins. In Game 6, the Attacker, playing Black, mated in 22 plies. The Materialist’s evaluation has no king-safety term, so it cannot see that its king is under pressure. The Attacker rewards pieces near the enemy king, a broken pawn shield and checks, so it moves toward positions where a mate may occur.

The Materialist only counts material. Its wins, in games 2, 5 and 9, took 361, 208 and 204 plies. The Materialist won only 1 of its 5 games as White and the Attacker won 4 of 5 as Black. The strategy mattered more than the first move. In the games the Attacker lost (5 and 9, both as White), its aggressive play may have overextended and given the Materialist material to collect. That is the main risk of the Attacker’s strategy.